BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

CPU Architecture and the Fetch-Execute Cycle

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is a logic error?

2. Which description matches an AND gate symbol?

A) Triangle with output circle
B) Flat input; rounded output
C) Curved input; pointed output
D) A decision diamond

3. "Iterative testing happens only after the entire finished product is delivered." Correct this. Explain why.

4. A booking program is tested before release. "Final testing means no testing is needed while building it." Correct this. Why are both needed?

Today's objectives

Explain the core ideas of cpu architecture and the fetch-execute cycle using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

ALU
Performs arithmetic and logical operations.

Control Unit
Coordinates instruction execution and the CPU's internal operations.

Registers
PC stores next instruction address, MAR stores address accessed, MDR holds transferred data/instruction and ACC holds calculation results.

Fetch-decode-execute
Instruction fetches from memory, decodes, executes and the cycle repeats.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

ALU	Performs arithmetic and logical operations.
Control Unit	Coordinates instruction execution and the CPU's internal operations.
Registers	PC stores next instruction address, MAR stores address accessed, MDR holds transferred data/instruction and ACC holds calculation results.
Fetch-decode-execute	Instruction fetches from memory, decodes, executes and the cycle repeats.

I Do - One worked example

Trace how the PC, MAR and MDR participate in fetching an instruction.

1. PC holds the address of the next instruction.
2. The address is copied to MAR.
3. The instruction at that address is transferred to MDR.
4. The instruction is decoded/executed, and PC is updated as appropriate.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Explain how the ALU and accumulator work together for an addition.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the final value of the illustrative accumulator variable.

```
acc = 4
acc = acc + 6
print(acc)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State the purpose of MAR and MDR. [4 marks]

Q3. What does the Program Counter hold? [1 mark]

Q4. Explain the role of the Control Unit. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. The MDR stores the address of the next instruction. ☐ T ☐ F
2. Performs arithmetic and logical operations. ☐ T ☐ F
3. Coordinates instruction execution and the CPU's internal operations. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A mistake in the algorithm that gives the wrong result, even though the code follows the language's rules. ☐

Q2 B - Flat input; rounded output ☐

Q3 Testing and correcting parts during development as the solution is built. ☐

Q4 Final testing checks the finished program against its requirements. Testing during development finds faults earlier. Both are needed. ☐

Code First - Model

```
acc = 4
acc = acc + 6
print(acc)
```

Output: 10

We Do - Check the model

1. Control Unit coordinates execution of the instruction. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 MAR: address of memory location accessed; MDR: data/instruction being transferred. ☐

Q3 Address of the next instruction to fetch. ☐

Q4 Coordinates CPU operations and controls execution of instructions. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: